1841

Swiss physiologist Albert von Kölliker showed that, like every other living cell, each sperm is a single cell with its own nucleus.

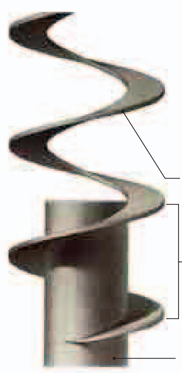
1-3 MILLIMETERS PER MINUTE THE SPEED A SPERMATOZOAN CAN TRAVEL

IN 1841, GERMAN PHYSICIAN JULIUS VON MAYER (1814–78) first proposed that **energy could never be created or destroyed**. This is now known as the first law of thermodynamics (see 1847–48). Von Mayer also proposed that work and heat are equivalent—a certain amount of work will always produce a certain amount of heat—as English physicist James Joule (1818–89) discovered independently two years later. However, it was some time before either von Mayer or Joule's work was acknowledged.

In contrast, improvements made by Swiss physiologist **Albert von Kölliker** (1817–1905) to microscope techniques, such as the staining of samples, were soon acknowledged and adopted. Von Kölliker also confirmed that **each sperm and egg is a cell with a nucleus**, adding to the emerging science of **histology**—the science of living cells.

In the UK, engineer **Joseph Whitworth** (1803–87) identified and solved a basic problem with assembling precision machines—the variation in screws. Whitworth devised a system of **standardization for screw threads and pitches**. When several railroad companies decided to adopt it, other organizations quickly followed suit. This system is now known as the BSW (British Standard Whitworth) system.

In Antarctica, British explorer **James Clark Ross** (1800–62) discovered and named the Victoria Barrier, later known as the Ross Ice Shelf.



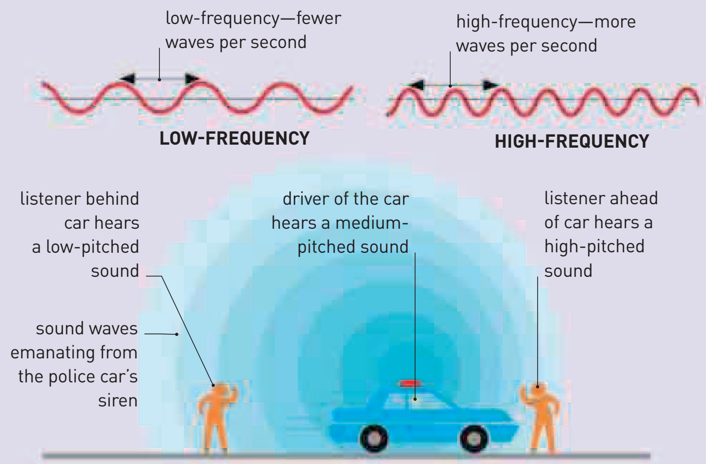
Screw threads
Joseph Whitworth set a standard for screw threads with a fixed angle of 55° for the thread.

1842–1843

British engineer James Nasmyth's steam hammer was able to forge large pieces of wrought iron and was developed in response to the increased demands of 19th-century engineering.

THE DOPPLER EFFECT

The sound of a police siren approaching rises in pitch as the sound waves are squashed in front of the moving car. As the car moves past and away from the listener, the pitch drops as sound waves stretch out in its wake. This occurs because, as the police car approaches, each successive sound wave begins closer to the car. As the police car moves away, each successive sound wave starts farther away from the car.

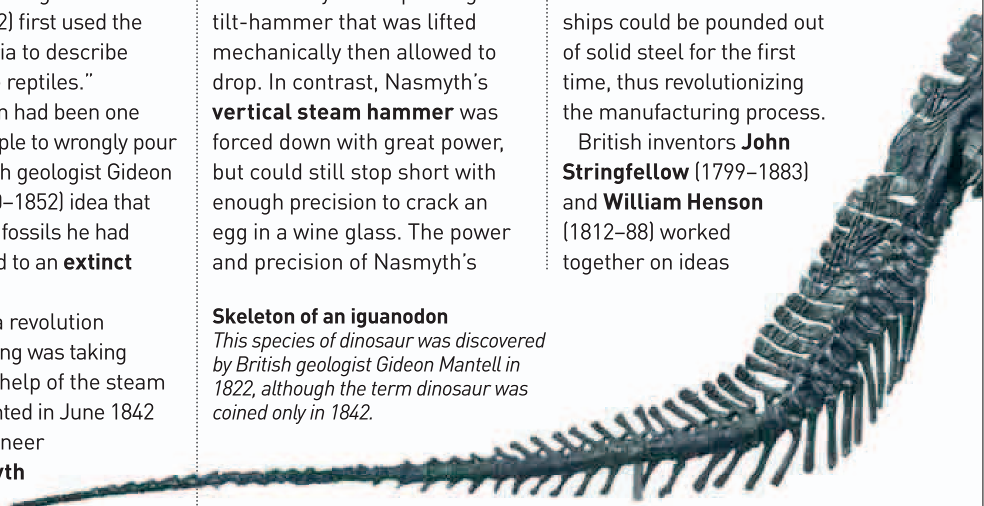


IN 1842, 25 YEARS AFTER THE FIRST RECOGNIZED FOSSIL was found, British biologist **Richard Owen** (1804–92) first used the word dinosauria to describe these "terrible reptiles." Ironically, Owen had been one of the first people to wrongly pour scorn on British geologist Gideon Mantell's (1790–1852) idea that the Iguanadon fossils he had found belonged to an **extinct giant reptile**.

Meanwhile, a revolution in manufacturing was taking place with the help of the steam hammer, patented in June 1842 by British engineer **James Nasmyth** (1808–90).

Previously, iron foundries had shaped iron weakly and inaccurately with a pivoting tilt-hammer that was lifted mechanically then allowed to drop. In contrast, Nasmyth's **vertical steam hammer** was forced down with great power, but could still stop short with enough precision to crack an egg in a wine glass. The power and precision of Nasmyth's steam hammer meant that such things as railroad wheels and the first steel hulls for ships could be pounded out of solid steel for the first time, thus revolutionizing the manufacturing process.

British inventors **John Stringfellow** (1799–1883) and **William Henson** (1812–88) worked together on ideas



Skeleton of an iguanodon
This species of dinosaur was discovered by British geologist Gideon Mantell in 1822, although the term dinosaur was coined only in 1842.

January 27 James Clark Ross discovers the active volcano Mount Erebus in Antarctica

June Julius von Mayer shows that **work and heat are equivalent**

Joseph Whitworth introduces a **standard screw system**

Albrecht von Kölliker shows that **sperm are cells**

January 1842 American medical student William E. Clarke makes the first dental extraction under general anaesthetic

1842 Richard Owen coins the name **dinosauria**

March 30, 1842 American physician Crawford Long makes the **first surgical operation under general anaesthetic**

June 1842 **James Nasmyth** patents the steam hammer

1842 Christopher Doppler discovers what became known as **the Doppler effect**